

Problems you didn't know you probably don't need

Excessive exercises aimed for foundational
conceptual understanding building

Lim Yun Zhe

Forewords

The *current version* should cover some of the concepts needed in SPM except for data analysis and financial mathematics, especially those that may be useful for further math. Hopefully this helps someone.

The various topics are arranged into categories of approximate fields of mathematics, but not necessarily by topic, and are some problems may be labeled with asterisks to indicate additional difficulty, which is approximately rated by absolute difficulty (where Complex $\neq$ Difficulty) but difficulty may vary from person to person. The difficulty are as the example below.

- * 1. This is a slightly hard problem.
- ** 2. This is a hard problem.
- *** 3. This is a very hard problem.

Contents

1	Algebra	2
2	Geometry	5
3	Calculus	8
4	Combinatorics	11

1 Algebra

1. The graph of $x^2 + y^2 = 1$ is the unit circle for $x, y \in \mathbb{R}$. Does it admit an function $f : \mathbb{R} \rightarrow \mathbb{R}$ by mapping x to y ? Explain.
2. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = |x - |x - 2||$.
 - (a) Sketch the graph by plotting integer points
 - (b) State the domain, codomain, the range, and type of relation of the function.
3. Let $g : \mathbb{Q} \rightarrow \mathbb{Q}$ be a quadratic function such that $g(1) = 3$, $g(-1) = 24$ and $g(3) = 57$. ($\mathbb{Q}$ is the set of rationals.)
 - (a) Determine the function $g(x)$.
 - (b) Does g have a value of x that maps to itself? State it, or explain why not so.
4. Consider the function $f : \mathbb{R} \setminus \left\{ -\frac{d}{c} \right\} \rightarrow \mathbb{R}$ with $f(z) = \frac{az + b}{cz + d}$, where a, b, c, d are constants with $ad - bc \neq 0$. Also, let $\mathfrak{H} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.
 - (a) Calculate $f^n(z)$ and A^n for $n = 2$, do you see any coincidence? (If not, calculate for $n = 3$ as well.)
 - (b) Consider $\mathfrak{H} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}$, find $f^n(z)$ for any n .
5. Consider functions $f(x) = x^3 + 1$ and $g(x) = \frac{1}{x} - 1$.
 - (a) Find fg and gf .
 - (b) For which values are the functions above not defined for?
6. Determine if the following functions have inverses. If no, explain why, else, calculate the inverse function.
 - (a) $f : \mathbb{R} \rightarrow \mathbb{Z}$, where $f(x) = \lfloor x \rfloor$. ($\lfloor x \rfloor$ is the largest integer not larger than x .)
 - (b) $g : \mathbb{R}^+ \rightarrow \mathbb{R}$, where $g(x) = \log x$.
 - (c) $h : \mathbb{Q} \rightarrow \mathbb{Q}$, where $h(x) = x^3$.
- * 7. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ where $g(g(x)) = x$ for all real $x \in \mathbb{R}$. Show that $g(x)$ must have an inverse.
8. Given that α and β are the roots of the quadratic $x^2 - 5x + 1$. Construct a quadratic equation that has the following roots.
 - (a) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$.
 - (b) $\sqrt{1 + \alpha^2}$ and $\sqrt{1 + \beta^2}$.
 - (c) $k\alpha + \beta$ and $\alpha + k\beta$, where k is an constant.
9. Let α be a root of $x^2 - x + 1$. For all positive integer $n \geq 2$, express α^n in the form of $p_n\alpha + q_n$, where p_n and q_n are integers.

10. By completing the square, or otherwise, show the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}}$$

yields the root for the quadratic $ax^2 + bx + c$.

11. Consider quadratics $f_k(x) = a_kx^2 + b_kx + c_k$ for $k = 1$ and $k = 2$, where a_k , b_k , and c_k are constants. Determine the criteria when $y = f_1(x)$ and $y = f_2(x)$ will meet, and the number of incident points for the conditions.

12. By completing the square, solve the inequality $3x^4 < 12x^2 - 7$.

*

13. Alice and Bob plays a game. Alice writes a quadratic $ax^2 + bx + c$ on a whiteboard using a black marker, and another quadratic $px^2 + qx + r$ using a red marker. At each round, Bob can perform one of the following actions on the black quadratic.

- Swap the places of a and c .
- Replace b with $-b$.
- Replace the x into some $x + k$ and rewrite the final quadratic instead.

Bob wins if he can transform the black polynomial to the red polynomial. Else, Alice wins.

- (a) On a game, Alice wrote the quadratic $x^2 + 5x + 2$ using a black marker, and writes the quadratic $3x^2 + 2x + 1$ using the red marker. Can Bob win?
- (b) Prove that Alice have a winning strategy.

14. Solve the following simultaneous equation.

$$\begin{cases} x - y = 2 \\ xy - 3y + 1 = 0 \end{cases}$$

15. It is given that $x^2 - y^2 = 3$ and $x^2 + 3xy + 2y^2 = 12$.

- (a) Show that $(x + y)^2 = 9$.
- (b) Hence, or otherwise, solve for x and y .

16. Show that the following simultaneous equation have infinite solutions.

$$\begin{cases} x + y + z = 2 \\ x - y + 2z = 4 \\ 3x + y + 4z = 8 \end{cases}$$

17. Solve the following simultaneous equation.

$$\begin{cases} x + \frac{1}{y} = 2 \\ y + \frac{1}{z} = 3 \\ z + \frac{1}{x} = 4 \end{cases}$$

- * 18. Given that a 3×3 matrix

$$A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

has determinant $\det(A) = aei + bfg + cdh - ceg - bdi - afh$. Then, the inverse of A can be written as

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} +(ei - fh) & -(di - fg) & +(dh - eg) \\ -(bi - ch) & +(ai - cg) & -(ah - bg) \\ +(bf - ce) & -(af - cd) & +(ae - bd) \end{pmatrix}.$$

(Hopefully you can see the pattern!) Then, a system of equation can be written as

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

with the normal form $A\mathbf{x} = \mathbf{b}$.

- When is there a unique solution to a three variable linear simultaneous equation, as by above?
- When there is not a unique solution, what determines if it have infinite solution or no solution? (Tips: Consider the rows of the matrix.)
- Define A_i by replacing the i -th row of the matrix A with $\mathbf{b}$. Show that the solutions can be found by

$$x = \frac{\det(A_1)}{\det(A)}, y = \frac{\det(A_2)}{\det(A)}, z = \frac{\det(A_3)}{\det(A)}.$$

19. Simplify and evaluate $\log_5 \left(\frac{\sqrt{9+4\sqrt{5}} + \sqrt{9-4\sqrt{5}}}{2} \right)$.

20. Solve the equation $\log_4(x^2 - 4) - \log_2(x - 2) = 1$.

21. Let $x_1, x_2, \dots, x_n$ be positive real numbers with $x_i \neq 1$ for all i . Show that

$$\frac{1}{1 + \log_{x_1} x_2 + \dots + \log_{x_1} x_n} + \frac{1}{\log_{x_2} x_1 + 1 + \dots + \log_{x_2} x_n} + \dots + \frac{1}{\log_{x_n} x_1 + \log_{x_n} x_2 + \dots + 1} = 1.$$

22. Simplify the following expressions.

(a) $\frac{1}{1 + \sqrt[3]{2} + \sqrt[2]{4}}$

(b) $\frac{1}{1 - \sqrt[5]{3}}$

23. Show that $3^{6n} + 9^{3n+1} - 27^{2n-1}$ is always divisible by 269 for all positive integer n .

24. Show that 8 always divides $(2^n + 3^n + 6^n)^2 - 1$ for any nonnegative integer n .

25. Let $T_1, T_2, \dots$ be an arithmetic progression where T_1, T_2, T_4 forms a geometric progression with $T_1 = e$.
- Show that $T_1 T_4 = T_2^2$.
 - Show that either $T_1 = d$, where d is the common difference, or $d = 0$.
 - For each case, find the general formula for T_n and the sum of the first n terms S_n .
26. Let $T_1, T_2, \dots$ be a geometric progression with each term a positive real. Known that the first three terms are an arithmetic progression, with $T_1 = \pi$,
- Show that $\log T_1, \log T_2, \dots$ is an arithmetic progression.
 - Show that the sequence is constant, i.e. $T_n = \pi$ for all n .
 - Find the sum of the first n terms, S_n .
27. Given that a sequence $\{T_n\}$ have series $\{S_n\}$, i.e. the sum of the first n terms.
- Show that if T_n is an arithmetic sequence, then S_n is a quadratic.
 - Show that if $\log S_n$ is linear, then T_n is a geometric sequence.
 - Given that $T_{n+1} S_n = T_n S_{n+1}$, determine the general formula for the sequence $\{T_n\}$ and its series $\{S_n\}$.
28. Consider the following relations, and transforming them into linear relations. (Constants are unknown to you, while other variables you have data for.)
- $s = v_0 t + \frac{1}{2} a t^2$, where v_0, a are constants.
 - $f = \frac{uv}{u+v}$, where f is constant.
 - $N = N_0 e^{-\lambda t}$, where N_0, λ are constants.
 - $Ax^2 + Bxy + Cy^2 = 0$, where A, B , are constants when known $C = 1$.

2 Geometry

- Given that a line has general equation $Ax + By + C = 0$, show that two points determine a line. (Clearly A, B, C not all 0 right?)
- Show the formula for the interior divisor of a line segment, where if $A(x_1, y_1)$ and $B(x_2, y_2)$ have an internal divisor P on AB such that $AP : PB = m : n$, then

$$P = \left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n} \right).$$

(Note: You may try solving in both analytical geometry by similarity or by vectors.)

3. Define the gradient of a line with equation $Ax + By + C = 0$ as $-\frac{A}{B}$, where we can actually generalise that the gradient is ∞ when $B = 0$, and define $\infty \times 0 = -1$ and nothing else.
 - (a) Show that under the definition above, if two lines are parallel, which never meet or coincide, with gradients m_1 and m_2 , then $m_1 = m_2$.
 - (b) Show that under the definition above, if two lines are perpendicular, then $m_1 m_2 = -1$.

(For a deeper dive or et cetera, actually look up projective planes, where we homogenize to $Ax + By + Cz = 0$ instead.)
4. Calculate the area of the convex polygon (convex hull of points) bounded by the points $A(-8, 9)$, $B(-9, 5)$, $C(-4, 9)$, $D(-6, 10)$, and $E(-6, 4)$. (Remember to check the orientation and order of points!)
5. For two points A, B on the plane, show that the locus of the point P , where $AP : PB = m : n$ for some constants m, n is a circle. (It is known as the Apollonius Circle!)
6. Two vectors $\vec{p}, \vec{q}$ are said to be independent if $a\vec{p} + b\vec{q} = \vec{0}$ implies $a = b = 0$. Show that two vectors are neither parallel nor anti-parallel (facing opposite directions) if and only if they are independent.
7. Consider a triangle ABC on the plane. Show that no matter where we place O , then for a fixed point P , there is a unique representation $(x : y : z)$ with $x + y + z = 1$ of the point P where $\vec{OP} = x\vec{OA} + y\vec{OB} + z\vec{OC}$. (i.e. We can find such a unique $(x : y : z)$, and it doesn't change even if we move O .)
8. Consider a triangle ABC with circumcircle ω , circumcenter O and circumradius R . Then, consider extending AO to meet ω again at A' . By considering $\triangle AA'B$ and by similar constructions, prove the sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

9. Show that if given $\angle BAC$ acute, and $BC < AB$, we can construct two such triangles $\triangle ABC$, unless $\sin \angle BAC = \frac{BC}{AB}$.
10. Consider a triangle $\triangle ABC$ where the foot of perpendicular of A to BC , is within the segment BC , and let it be D . Then, show the cosine rule,

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

11. Show the area of the triangle is given by

$$\frac{1}{2}ab \sin C.$$

12. By using the law of cosines and the area of a triangle by sines, show the Heron's formula, where the area of the triangle is given by

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

where s is the semiperimeter of the triangle, $2s = a + b + c$.

13. Convert the following from degrees to radian, or from radian to degrees.

- (a) 4π rad
- (b) 3060°
- (c) $45^\circ 15' 12''$
- (d) $\frac{5\pi}{4}$ rad
- (e) 120 rad
- (f) 45
- (g) π°

14. Show that for a segment (region bounded by arc and chord) with the angle subtended to the center as θ ,

- (a) The perimeter of the segment is

$$r \left(\theta + 2 \sin \frac{\theta}{2} \right).$$

- (b) The area of the segment is

$$\frac{1}{2} r^2 (\theta - \sin \theta)$$

for $0 < \theta < \pi$ and

$$\frac{1}{2} r^2 (\theta + \sin \theta)$$

for $\pi < \theta < 2\pi$.

15. Sketch the graph of

- (a) $y = \tan x$,
- (b) $y = 4 \sin 2x$,
- (c) $y = |\frac{1}{2} \sin x - 1|$,
- (d) $y = ||\sin 2x| - \frac{1}{2}|$.

16. For a triangle $\triangle ABC$, show that

- (a) $\tan A + \tan B + \tan C = \tan(A + B + C)$,
- (b) $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \tan(\frac{A+B+C}{2})$,
- (c) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$,
- (d) $\cos^2 A + \cos^2 B + \cos^2 C + 2 \cos A \cos B \cos C = 1$.

17. Show that

(a) $\tan^{-1} \frac{1}{2} = \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7},$

(b) $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8},$

(c) $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi.$

18. Show that

(a) the general solution of $\sin \theta = a$ is

$$\theta = n\pi + (-1)^n \sin^{-1} a,$$

(b) the general solution of $\cos \theta = b$ is

$$\theta = 2n\pi \pm \cos^{-1} b,$$

(c) the general solution of $\tan \theta = c$ is

$$\theta = n\pi + \tan^{-1} c.$$

3 Calculus

1. Find the value of the following.

(a) $\lim_{x \rightarrow 0} \frac{x+1}{x^2+1}$

(b) $\lim_{p \rightarrow 0} \frac{x^2 - 5x + 6}{x^2 - 3x + 2}$

(c) $\lim_{x \rightarrow -\infty} \frac{8}{x^2}$

(d) $\lim_{x \rightarrow \infty} \frac{x^2 + 2}{2x^2 + x + 1}$

(e) $\lim_{n \rightarrow \infty} \left(\frac{3}{4}\right)^n$

(f) $\lim_{x \rightarrow 3} \frac{\sqrt{2x+3} - 3}{x - 3}$

(g) $\lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h}$

2. By using $\frac{df}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$, calculate $\frac{dy}{dx}$ when

(a) $y = x^2 + 3$

(b) $y = -\frac{1}{x^2} + 2$

(c) $y = x^n, n \in \mathbb{Z}.$

3. By using the basic definition, show that for functions $p(x)$, $q(x)$, and for constants a , b , we have

$$\frac{d}{dx}[ap(x) + bq(x)] = a\frac{d}{dx}p(x) + b\frac{d}{dx}q(x).$$

4. Find the first (and second) derivatives of the following functions with respect to x .

- (a) $f(x) = \frac{4}{x}$
- (b) $p(x) = 3x^3 + 7$
- (c) $q(x) = \frac{x^3+1}{x^2}$
- (d) $f(x) = 4x^2(\sqrt[3]{x} - x)$
- (e) $g(x) = (4 + x)^7$
- (f) $f(x) = 2\sqrt{1 - tx^2}$
- (g) $g(x) = (x + 1)^4(x - 1)^5$
- (h) $h(x) = \frac{4-3x}{\sqrt[3]{2-x}}$
- (i) $s(x) = x^{\frac{2}{3}}(d^2 - x^2)^{\frac{1}{6}}$
- (j) $f(x) = [g(x)]^2 - g^2(x)$, $g(x) = x^4 - 1$

5. Prove that for a quadratic curve, no line can be both a normal and a tangent to the curve. (Solution using algebra methods exists as well!)

*

6. Given a curve $y = ax^3 + bx + c$ having a tangent passing through $(0, -1)$. Show that regardless of how a , b changes for fixed c , this tangent always intersect at another point with $x = x_0$, and $\frac{x_0}{\sqrt[3]{a}}$ is constant. (You may show that the intersection must have the same x_0 for b fixed, then show that the intersection have $x_0/\sqrt[3]{a}$ fixed.)

7. Determine the turning points of $y = x^5 - 4x^3 - 9x + 12$, and determine if they are the local maximum or the minimum.

8. Consider $f(x) = (x^2 + 1)(x + 1)(x - 3)(x - 4) - 12$. Determine for which $n \in \mathbb{Z}$ is $f(n) = 0$, $f(n) > 0$, and $f(n) < 0$.

*

9. Given that $\frac{d}{dx} \cos x = \sin x$ and $\frac{d}{dx} e^x = e^x$.

- (a) Calculate $\frac{d}{dx} \sin x$, then demonstrate that $y = A \sin(\omega x) + B \cos(\omega x)$ satisfies the equation $\frac{d^2 y}{dx^2} + y = 0$.
- (b) Then, show that $e^{it} = \cos t + i \sin t$, where i is the imaginary unit with $i^2 = -1$.

10. Given that $\frac{d}{dx} \ln x = \frac{1}{x}$,

- (a) Calculate $\frac{d}{dx} x \ln x$.
- (b) Then, find $\int \ln(ax + b) dx$.

11. (a) Show that

$$\frac{d}{dx} \frac{(x+2)^3(x^2-4)^4}{x^2+4} = \frac{(x-2)^3(x+2)^6(9x^3-6x^2+52x-24)}{(x^2+4)^2}.$$

- (b) Hence, or otherwise, calculate

$$\int \frac{(x-2)^3(x+2)^6(9x^3-6x^2+52x-24)}{(x^2+4)^2} dx.$$

- (c) Calculate

$$\int_{-1}^1 \frac{(x-2)^3(x+2)^6(9x^3-6x^2+52x-24)}{(x^2+4)^2} dx.$$

12. Evaluate the following integrals.

(a) $\int (6x^2 + 3x + 2) dx$

(b) $\int \sqrt[5]{x+2} dx$

(c) $\int (x^2 - 3x + 1)(2x + 3) dx$

(d) $\int_0^1 (7x - 6)^{23} dy$

(e) $\int_2^3 (7x - 6)^{23} dx$

13. By using the substitution $u = 6x^2 + 3x - 2$, calculate $\int (6x^2 + 3x - 2)^5(4x + 1) dx$.

14. If $\int_a^b f(x) dx = 6$ with $a \neq b$, can we calculate $\int_b^a 3f(2x) dx$? If so, calculate the value, or else, explain why.

15. Calculate the area bounded by these curves.

(a) $x = 1$, $x = 3$, x -axis, and $y = x^3$.

(b) x -axis, and $y = x^4 - 2x$.

(c) x -axis, y -axis, and $y = \sqrt{1 - 5x}$.

(d) x -axis, y -axis, and $y = -\frac{4x+3}{x+2}$.

(e) negative x -axis, positive y -axis, and $y = (x+1)(x+2)(x+3)$.

(f) $x + 2y = 22$, $y = (x+4)^3$, and x -axis.

(g) $y = \sqrt{x}$, $y = 1/x$, and $180y + 13x^2 - 125x - 68 = 0$.

16. Calculate the volume of when the region bounded by the x -axis and $y = -x(x-4)$ is rotated through 360° around the x axis.

4 Combinatorics

- Given six books to be distributed to three persons, A , B and C , how many ways are there if
 - there are no restrictions,
 - each person must get an equal amount of books,
 - each person must get a different nonzero amount of books.
- Let A , B , C be a triangle on a plane. If there are a points between B and C , b points between A and C , and c points between A and B , and we can only choose from these $a + b + c$ points, how many non-degenerate triangles can be constructed?
- Given r identical red balls, y identical yellow balls, and b identical blue balls, how many ways are there to distribute between two persons, Alice and Bob, if
 - there are no restrictions,
 - each person gets all colours of balls,
 - each person gets balls of at most two colours.
- Determine the number of five digit numbers divisible by 45 and containing on the digits 1, 2, 3, 4, 5, possibly multiple times.
- Given a deck of cards (52 cards), what is the probability of drawing four cards, where we can form two pairs based on similarities (same suit or same value)?
- Find the probability that when a fair 6 faced ordinary dice (D6) is thrown thrice, the sum of values is
 - 3,
 - 12,
 - at least 12.
- By using the binomial formula or otherwise, show that if $X \sim B(n, p)$, then

$$\mathbb{P}(X = r) = \binom{n}{r} p^r (1 - p)^{n-r}.$$

Then, by using the definition for variance, show that $\sigma = \sqrt{np(1 - p)}$.

- By using the fact that the PDF (probability density function) of $X \sim N(\mu, \sigma^2)$ is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right),$$

where $\exp x = e^x$, show that $Z = \frac{X - \mu}{\sigma}$ standardises the normal distribution, or $Z \sim N(0, 1)$.